# Dr. Ayan Chatterjee

chatterjee.ay@northeastern.edu, bioclarityai@gmail.com,
chatterjee.ayan@itrakneuro.com
177 Huntington Avenue, Boston, MA 02115
Contact: +1 (617) 840-8467
Personal website
LinkedIn


| | |
|---|---|
| Research Interests | Artificial General Intelligence (AGI), Drug Discovery, Link Prediction, Graph Machine Learning (ML), Generalizability, Social Networks, Biological Networks. |

**Current activities**

Founded **BioClarity AI**

- BioClarity AI is a Massachusetts-based single member-LLC that provides AI solutions to pharmaceutical companies (e.g., Alexion AstraZeneca Rare Disease).
- BioClarity has created ML models for protein-peptide interaction prediction, which has helped develop a targeted AAV capsid in AstraZeneca's gene therapy efforts. The prediction has gained success in experiments.
- BioClarity has also gained success in deploying AAV capsid packability prediction models that have gained experimental success.

Co-founded **ITrakNeuro Inc**.

- ITrakNeuro is a New York-based startup focusing on developing AI-powered early diagnostic tools for cancer.
- ITrakeNuro has developed AI products for identifying brain cancer using vision-based ML models which are being deployed at Banner Health, Arizona, and Manipal Hospitals, India.
- ITrakNeuro was co-founded with
    - Prof. Janet Paluh from the University at Albany,
    - Dr. Amitava Mukherjee who is a professor at Amrita University, India and an ex-IBM executive, and
    - Rounak Chatterjee from the University of Birmingham, UK.

| | |
|---|---|
| Education | **Ph.D. in Network Science (with Computer Science specialization), 2019-2024**<br>Network Science Institute, Northeastern University, Boston, MA.<br>Advisor: Prof. Tina Eliassi-Rad, President Joesph E. Aoun Professor, Khoury College of Computer Sciences and Network Science Institute, Northeastern University, Boston, MA.<br>Dissertation: Enhancing Generalizability in Static and Temporal Link Prediction with Applications in Drug Discovery<br>GPA: 4.0/4.0<br>Committee: Tina Eliassi-Rad, Samuel Scarpino, Olga Vitek, and John Palowitch<br><br>**Master of Technology (M.Tech) in Electronic Systems Engineering, 2015-2017**<br>Indian Institute of Science, Bangalore, India<br>Thesis Topic: Multipacket Reception in Wireless Networks<br>GPA: 6.5/8.0<br>Advisor: Dr. Chandramani Singh and Dr. T . V. Prabhakar<br><br>**Bachelor of Electronics Telecommunication Engineering, 2011-2015**<br>Jadavpur University, Kolkata, India<br>Thesis Topic: FPGA-based VGA Controller Design for CRTs and Graphics Processor Design<br>GPA: 9.78/10<br>Advisor: Prof. M. K. Naskar |
| Professional Experience and Internships | **Research Assistant, Network Science Institute, Northeastern University, Boston, Fall 2019 - 2024**<br>Topic: Generalizability in Graph Machine Learning<br>Advisor: Prof. Tina Eliassi-Rad<br><br>**Internship at Google DeepMind, Fall 2023 - Spring 2024**<br>Topic: Foundation Model for Temporal Link Prediction<br>Mentor: John Plaowitch, Google DeepMind (previously, Google Research)<br><br>**Internship at Alexion AstraZeneca Rare Disease, Summer 2023**<br>Topic: Protein-protein interaction prediction using graph machine learning<br>Mentor: Babak Ravandi<br><br>**Computer Architect at NVIDIA, 2017-2019**<br>Topic: Designing low-power GPU architecture<br>Manager: Raghavendra H. Bhat<br><br>**Internship at IBM Global Remote Monitoring Program, August 2015 - February 2015**<br>Topic: Control of complex networks<br>Advisor: Dr. Amitava Mukherjee |

**Internship at Indian Statistical Institute, Kolkata, India. May 2014 - July 2014**
Topic: Software implementation of several state-of-the-art stream ciphers
Advisor: Prof. Subhamoy Maitra

Peer-reviewed Publications

**Machine Learning in Biology, Drug Discovery, and Cancer Research**

**Topology-Driven Negative Sampling Enhances Generalizability in Protein-Protein Interaction Prediction.** Chatterjee, A., Ravandi, B., Haddadi, P ., Philip, N. H., Abdelmessih, M., Mowrey, W. R., Ricchiuto, P ., Liang, Y ., Ding, W., Mobarec, J. C., & Eliassi-Rad, T . (2024). (Accepted for publication at Oxford Bioinformatics) Pre-print: https://doi.org/10.1101/2024.04.27.591478.

**SIENNA: Lightweight Generalizable Machine Learning Platform for Brain Tumor Diagnostics**. S. Sunil, R. S. Rajeev, A. Chatterjee, J. Pilitsis, A. Mukherjee, J. L. Paluh. Manuscript under preparation for submission to the American Journal of Roentgenology. Pre-print: https://www.medrxiv.org/content/10.1101/2024.04.03.24305210v1.

**Representation Learning of Human Disease Mechanisms for a Foundation Model in Rare and Common Diseases**. Babak Ravandi, William R. Mowrey, Ayan Chatterjee, Piero Ricchiuto, Parham Haddadi, Mario Abdelmessih, Wei Ding, Simon Lambden, Michaël Ughetto, Ian Barrett, Tom Diethe, Guillermo Del Angel, and Tina Eliassi-Rad. Manuscript under preparation for submission to Cell Patterns. Pre-print: https://www.biorxiv.org/content/10.1101/2024.11.19.624381v1.

**Improving the generalizability of protein-ligand binding predictions with AI-Bind**. A. Chatterjee, R. Walters, Z. Shafi et al. *Nature Communications* 14, 1989 (2023). https://doi.org/10.1038/s41467-023-37572-z.

**SCENIC: An Area and Energy-Efficient CNN-based Hardware Accelerator for Discernable Classification of Brain Pathologies using MRI**. B. S. T . Naidu et al. 2022 35th International Conference on VLSI Design and 2022 21st International Conference on Embedded Systems (VLSID), Bangalore, India, 2022, pp. 168-173, doi: 10.1109/VLSID2022.2022.00042.

**MilkyBase, a database of human milk composition as a function of maternal-, infant- and measurement conditions.** Pacza, T ., Martins, M.L., Rockaya, M. et al. Sci Data 9, 557 (2022). https://doi.org/10.1038/s41597-022-01663-1.

**Core Machine Learning**

**Transfer Learning for Temporal Link Prediction.** A. Chatterjee, B. Ikica, B. Ravandi, J. Palowitch. Submitted to the International workshop on Scalable and Deep Graph Learning and Mining (SGLM) in conjunction with International Joint Conference on Neural Networks (IJCNN2025).

**Explaining Node Embeddings.** Z. Shafi, A. Chatterjee, T. Eliassi-Rad. Accepted for publication in the Transactions on Machine Learning Research (2025). https://openreview.net/pdf?id=QQZ8uPxFb3

**Inductive Link Prediction in Static and Temporal Graphs for Isolated Nodes**. A. Chatterjee, R. Walters, G. Menichetti, T . Eliassi-Rad. Temporal Graph Learning Workshop @ NeurIPS 2023. https://openreview.net/pdf?id=DRrSYKNhD1.

**GRASP: Accelerating Shortest Path Attacks via Graph Attention**. Zohair Shafi, Benjamin A Miller, Ayan Chatterjee, Tina Eliassi-Rad, Rajmonda S. Caceres. Deep Learning on Graphs Workshop @ KDD 2023. https://arxiv.org/abs/2310.07980.

**Disentangling Node Attributes from Graph Topology for Improved Generalizability in Link Prediction**. A. Chatterjee, R. Walters, G. Menichetti, T . Eliassi-Rad. arXiv preprint https://arxiv.org/abs/2307.08877.

**Core Network Science and Wireless Networks**

**Deterministic random walk model in NetLogo and the identification of asymmetric saturation time in random graphs**. A. Chatterjee, Q. Cao, A. Sajadi, et al. Appl Netw Sci 8, 33 (2023). https://doi.org/10.1007/s41109-023-00559-2.

**Snickometer edge detection by feature extraction in TF plane and wavelet domain.** A Ghosh, A Chatterjee, A Roy, A Mukherjee, M.K. Naskar. 2018 IEEE Applied Signal Processing Conference (ASPCON), 143-148.

**Non-autonomous dynamic network model involving growth and decay**. A Chatterjee, A Chakraborty, S Pal, A Mukherjee, M.K. Naskar. 016 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS), Bangalore, India, 2016, pp. 1-6, doi: 10.1109/ANTS.2016.7947793.

**O(N^2) Heuristic for the Estimation of Driver Nodes for the Controllability of Directed Complex Networks**. A. Bhattacharya, A. Chatterjee, D. Das, A. Mukherjee, M.K. Naskar. LARGE SCALE COMPLEX NETWORK ANALYSIS: LSCNA2015, 32.

**AS 802.15. 4: A modified IEEE 802.15. 4 standard for more reliable communication and utilization of inactive period using optimized sleep period**. B. Bandyopadhyay, S.J. Ahmed, D. Das, A. Chatterjee. 2014 Annual IEEE India Conference (INDICON), 1-6.

**Characterizing behavior of Complex networks against perturbations and generation of Pseudo-random networks**. D. Das, A. Chatterjee, B. Bandyopadhyay, S.J. Ahmed. 2014 Annual IEEE India Conference (INDICON), 1-6.

**Markov chain-based analysis of IEEE 802.15. 6 mac protocol in real-life scenario**. B. Bandyopadhyay, D. Das, A. Chatterjee, S.J. Ahmed, A. Mukherjee. Proceedings of the 9th international conference on body area networks, 331-337.

**A Degree-first Greedy Search Algorithm for the Evaluation of Structural Controllability of Real World Directed Complex Networks**. D. Das, A. Chatterjee, N. Pal, A. Mukherjee, M.K. Naskar. Netw. Protoc. Algorithms 6 (1), 1-18.

**Heuristic for maximum matching in directed complex networks**. A. Chatterjee, D. Das, M. K. Naskar, N. Pal and A. Mukherjee. 2013 International Conference on Advances in Computing, Communications and Informatics (ICACCI), Mysore, India, 2013, pp. 1146-1151, doi: 10.1109/ICACCI.2013.6637339.

**Link Capacity Distributions and Optimal Capacities for Competent Network Performance**. S. Pal, A. Chatterjee, D. Bakshi, A. Mukherjee. arXiv preprint https://arxiv.org/abs/2002.02522.

Book Chapters

**Design of Structural Controllability for Complex Network Architecture**. A. Mukherjee, A. Chatterjee, et al. Advanced Methods for Complex Network Analysis, edited by Natarajan Meghanathan, IGI Global, 2016, pp. 98-124. https://doi.org/10.4018/978-1-4666-9964-9.ch004.

Talks and Presentations

**Protein-Protein Interactions and Negative Sampling**. Talk at AI Workshop at Alexion AstraZenecsa Rare Disease. March 2025.

**Toward a Foundation Model in Temporal Link Prediction**. Research Presentation at Google DeepMind. Summer 2024.

**Identifying interactions between novel protein targets and ligands: AI-Bind and AI-assisted molecular docking**. Khoury College of Computer Sciences, Northeastern University, Boston. 2022.

**Channing Methods Meeting on identifying interactions between novel protein targets and ligands: AI-Bind and AI-assisted molecular docking**. Harvard T .H. Chan School of Public Health, Harvard University, Boston. 2022.

Presented **Non-Autonomous Dynamic Network Model Involving Growth And Decay** at the IEEE International Conference on Advanced Networks and Telecommunications Systems between 6-9 November 2016 in Bangalore, India.

| | |
|---|---|
| Teaching and Mentoring | **Teaching assistant for Fall 2023: NETS 7332 -- Machine Learning with Graphs (CRN 19900)**. Course website: https://eliassi.org/fa23nets.html.<br><br>**Teaching assistant for Fall 2024: NETS 7332 -- Machine Learning with Graphs – CS 7332 (CRN 20227) & NETS 7332 (CRN 17372)**. Course website: https://eliassi.org/fa24nets.html.<br><br>Mentored **3 undergraduate and 3 master's students** at Northeastern University in 4 different research projects involving AI and biology (drug discovery and food nutrient exploration).<br><br>Mentored **1 intern and 1 new** hire at NVIDIA, Bangalore, India. |
| Academic Service | *Peer reviewer* for<br>● Journal of Machine Learning Research,<br>● Workshop on Graph Learning Benchmarks @ KDD 2023,<br>● Learning on Graphs Conference 2022/2023,<br>● Transactions on Knowledge Discovery from Data,<br>● Temporal Graph Learning Workshop @ NeurIPS 2023,<br>● Nature Communications journal.<br><br>*Program committee member* at<br>● ACM International Conference on Information and Knowledge Management (CIKM) 2024,<br>● ACM International Conference on Information and Knowledge Management (CIKM) 2025.<br><br>Undergraduate Student Representative at Jadavpur University, India. 2011-2015. |
| Other Academic Achievements and Special Laurels | Pharmacy Golden Jubilee Best Student of the Year award for the student who scored the highest percentage of marks in the UG Final Examination 2015.<br><br>University Medal for standing first in merit at the Bachelor of Electronics & Telecommunication Engineering (BETCE) Final Examination 2015.<br><br>J. P.. Saha Memorial Gold-Centered Silver Medal for securing the highest total marks in the theoretical papers at the Bachelor of Electronics & Telecommunication Engineering (BETCE) Final Examination 2015.<br><br>Pran Kumar Bhattacharya Memorial Gold Medal for standing first in order of merit at the Bachelor of Electronics & Telecommunication Engineering (BETCE) Final Examination 2015.<br><br>Prof. Sudhanshu Deb Memorial Gold-Centered Silver Medal for securing the highest aggregate marks in IC Technology and Design and VLSI Design at BETCE Final Examination 2015. |

Dr. B. C. Roy Memorial Gold-Centered Silver Medal for securing the highest aggregate score in all courses at the Bachelor of Engineering Examination 2015.

Sourav Banerjee Memorial Gold-Centered Silver Medal for standing First in order of merit at the Bachelor of Electronics & Telecommunication Engineering (BETCE) Final Examination 2015.

Prof. Debidas Mukhopadhyay Memorial Gold-Centered Silver Medal for securing the highest aggregate marks in Electron Devices 1&2 at the Bachelor of Electronics & Telecommunication Engineering (BETCE) Final Examination 2015.

DAAD-WISE Scholarship in 2014 for summer research internship at RWTH Aachen University, Germany.

Summer Research Fellowship organized by the Indian Academy of Sciences in 2014. Rank in West Bengal in Higher Secondary Examination (+2) among ~ 7,00,000 students in 2011.

54th Rank in the West Bengal Joint Entrance Examination (Engineering) among ~1,10,000 students in 2011.

7th Rank in the Secondary Examination (10) among ~ 9,00,000 students in 2009.

Mamraj Agarwal Rashtriya Puraskar for obtaining 93.25% in the Secondary Examination 2009 held under the West Bengal Board of Secondary Education.

**Outreach**

YouTube Channel for teaching Network Science and Graph Machine Learning to a broad audience: https://www.youtube.com/@NetworksAndGraphs.

**Extracurricular Activities and Hobbies**

Coordinator and Instructor at the Science Camp in May 2013, organized by the Matrix Publishers, Kolkata, India.

Guitarist in college band. Participated in various competitions representing Jadavpur University, India.

Guitar instructor at Forte Piano School of Music, Bangalore, India between 2016-2018.